Chapter 1. Real Numbers and Limits of Sequences

Field and order axioms, completeness, limits, monotone sequences, limsup and liminf, countability

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Analysis rests on a precise description of the real numbers. We take $\mathbb{R}$ as given by the field axioms, the order axioms and the completeness axiom, and derive everything else from them; Appendix A constructs a field satisfying the axioms. Book numbers are given in brackets.

1.1 Field and order axioms

Motivation. The algebraic and order rules of $\mathbb{R}$ are shared by $\mathbb{Q}$, so they cannot be what makes analysis work; still, every inequality manipulation later in the course reduces to them. We list them once and derive the familiar consequences, notably the triangle inequality.

Definition (Ordered field)

A **field** is a set F with operations $+$ and $\cdot$, both associative and commutative, with $\cdot$ distributive over $+$, identities $0 \neq 1$, additive inverses $-a$, and multiplicative inverses a^{-1} for $a \neq 0$ (axioms (F1)–(F9)). A field is **ordered** if there is a subset P (the positive elements) with (O1) $a, b \in P \Rightarrow a+b \in P$; (O2) $a, b \in P \Rightarrow ab \in P$; (O3) for each a exactly one of $a \in P$, $a = 0$, $-a \in P$ holds. We write $a < b$ (or $b > a$) if $b - a \in P$, and $a \leq b$ if $a < b$ or $a = b$.

Proposition 1.1 (1.1.1)

In a field:
(a) $-(-a) = a$, and $(a^{-1})^{-1} = a$ for $a \neq 0$;
(b) $a + b = a + c \Rightarrow b = c$, and $ab = ac$, $a \neq 0 \Rightarrow b = c$;
(c) $ab = 0 \iff a = 0$ or $b = 0$;
(d) $a \cdot 0 = 0$, $(-a)b = -(ab)$, $(-a)(-b) = ab$;
(e) the identities and inverses are unique.

Proof. (e) If $0, 0'$ are additive identities, $0 = 0 + 0' = 0'$; if b, b' are inverses of a, $b = b + (a + b') = (b + a) + b' = b'$; likewise for $\cdot$.

(b) Add $-a$ to both sides (multiply by a^{-1}).

(a) a and $-(-a)$ are both additive inverses of $-a$; use (e).

(d) $a \cdot 0 + a \cdot 0 = a(0 + 0) = a \cdot 0 = a \cdot 0 + 0$, so $a \cdot 0 = 0$ by (b); $ab + (-a)b = (a - a)b = 0$ gives $(-a)b = -(ab)$, and then $(-a)(-b) = -(a(-b)) = -(-(ab)) = ab$.

(c) If $ab = 0$ and $a \neq 0$, then $b = a^{-1}ab = 0$. ∎

Proposition 1.2 (1.1.2)

In an ordered field:
(a) $a \leq b$ and $b \leq a$ imply $a = b$;
(b) $a < b$, $b < c$ imply $a < c$;
(c) $a + b < a + c \iff b < c$;
(d) if $a > 0$ then $b < c \iff ab < ac$, and if $a < 0$ then $b < c \iff ab > ac$;
(e) $a^2 > 0$ for $a \neq 0$, in particular $1 > 0$;
(f) $a > 0 \Rightarrow a^{-1} > 0$, and $0 < a < b \Rightarrow 0 < b^{-1} < a^{-1}$;
(g) $a < b$, $c < d \Rightarrow a + c < b + d$;
(h) for $a, b \geq 0$: $a < b \iff a^2 < b^2$, and $a^2 = b^2 \Rightarrow a = b$.

Proof. (a) If $a \neq b$, both $b - a$ and $a - b = -(b - a)$ would be in P, against (O3).

(b) $c - a = (c - b) + (b - a) \in P$.

(c) $(a+c)-(a+b)=c-b$.

(d) $ac-ab=a(c-b)$; for $a>0$ use (O2) and trichotomy, for $a<0$ note $-a>0$.

(e) If $a>0$, $a^2\in P$; if $a<0$, $a^2=(-a)^2\in P$; and $1=1^2$.

(f) If $a^{-1}\le 0$ then $1=aa^{-1}\le 0$; multiply $a<b$ by $a^{-1}b^{-1}>0$.

(g) $(b+d)-(a+c)=(b-a)+(d-c)$.

(h) $b^2-a^2=(b-a)(b+a)$ and $b+a>0$ when $a,b\ge 0$ are not both 0. ■

$\mathbb{Q}$ and $\mathbb{R}$ are ordered fields. $\mathbb{C}$ is not: in an ordered field $-1<0<1=1^2$, but $i^2=-1$ would have to be positive. No finite field is ordered, since $1, 1+1, 1+1+1, \dots$ are strictly increasing.

The **absolute value** is $|a|=a$ for $a\ge 0$ and $|a|=-a$ for $a<0$.

Proposition 1.3 (1.1.3)

(a) $|a|\ge 0$, and $|a|=0 \iff a=0$;
(b) $|ab|=|a||b|$;
(c) for $b\ge 0$, $|a|\le b \iff -b\le a\le b$;
(d) $|a+b|\le |a|+|b|$ (triangle inequality);
(e) $||a|-|b||\le |a-b|$.

Proof. (a), (b) are case checks.

(c) If $a\ge 0$, $|a|\le b \iff a\le b$, and $-b\le 0\le a$ automatically; similarly for $a<0$.

(d) From $-|a|\le a\le |a|$ and the same for b, $-(|a|+|b|)\le a+b\le |a|+|b|$; apply (c).

(e) $|a|=|(a-b)+b|\le |a-b|+|b|$, and symmetrically. ■

1.2 The completeness axiom

Motivation. $\mathbb{Q}$ has a "hole" at $\sqrt{2}$: the set $\{x\in\mathbb{Q}: x^2<2\}$ is bounded but has no least upper bound in $\mathbb{Q}$. The completeness axiom says $\mathbb{R}$ has no such holes, and it is the single property behind every existence theorem in analysis: limits of monotone sequences, intermediate values, maxima of continuous functions.

$S\subset\mathbb{R}$ is **bounded above** if it has an **upper bound** u ($s\le u$ for all $s\in S$). A **least upper bound** (**supremum**) $\sup S$ is an upper bound smaller than every other upper bound; it is unique if it exists. Lower bounds and $\inf S$ are defined dually.

Definition (Completeness axiom)

(C) Every nonempty subset of $\mathbb{R}$ that is bounded above has a least upper bound.

Proposition 1.4 (1.2.1)

In an ordered field the following are equivalent:
(a) every nonempty set bounded above has a supremum;
(b) every nonempty set bounded below has an infimum.

Proof. (a)⇒(b): let $S\ne\emptyset$ be bounded below and L the set of lower bounds of S; $L\ne\emptyset$, and every $s\in S$ is an upper bound of L. Let $a=\sup L$. Each $s\in S$ is an upper bound of L, so $a\le s$: $a\in L$. If $b\in L$ then $b\le a$. So $a=\inf S$.

(b)⇒(a) is symmetric, or apply (b) to $-S = \{-s\}$. ■

A useful reformulation: $u = \sup S$ iff u is an upper bound and for every $\epsilon > 0$ there is $s \in S$ with $s > u - \epsilon$.

Proposition 1.5 (1.2.2)

$\mathbb{N}$ is not bounded above.

Proof. Otherwise $a = \sup \mathbb{N}$ exists; $a - 1$ is not an upper bound, so $a - 1 < n$ for some $n \in \mathbb{N}$, and then $n + 1 > a$ with $n + 1 \in \mathbb{N}$, a contradiction. ■

Corollary 1.6 (1.2.3)

$\inf\{1/n : n \in \mathbb{N}\} = 0$; that is, for every $\epsilon > 0$ there is $n \in \mathbb{N}$ with $1/n < \epsilon$.

Proof. 0 is a lower bound. If $\epsilon > 0$ were a lower bound, then $n \le 1/\epsilon$ for all n, contradicting Proposition 1.5. ■

Corollary 1.7 (1.2.4, Archimedean property)

For $a > 0$ and $b \in \mathbb{R}$ there is $n \in \mathbb{N}$ with $na > b$.

Proof. By Proposition 1.5 choose $n > b/a$. ■

Theorem 1.8 (1.2.5, density of the rationals)

If $a < b$ there is $r \in \mathbb{Q}$ with $a < r < b$.

Proof. First let $0 \le a < b$. Choose n with $1/n < b - a$ (Corollary 1.6). The set $\{m \in \mathbb{N} : m/n > a\}$ is nonempty (Corollary 1.7); let m be its least element. Then $(m-1)/n \le a$ (if $m = 1$ this holds as $a \ge 0$), so $a < m/n \le a + 1/n < b$. If $a < 0$, choose $k \in \mathbb{N}$ with $k > -a$, find r between $a + k \ge 0$ and $b + k$, and take $r - k$. ■

The irrationals are also dense: apply the theorem to $a/\sqrt{2} < b/\sqrt{2}$ (using Proposition 1.9).

Proposition 1.9 (1.2.6)

For each $a > 0$ there is a unique $b > 0$ with $b^2 = a$.

Proof. Uniqueness: Proposition 1.2(h).

Existence: $S = \{x > 0 : x^2 < a\}$ is nonempty (it contains $t = a/(1 + a)$, since $t < 1$ and $t < a$ give $t^2 < t < a$) and bounded above by $1 + a$ (if $x > 1 + a$ then $x^2 > x > a$). Let $b = \sup S > 0$. If $b^2 < a$, for $0 < h < 1$ we have $(b+h)^2 \le b^2 + h(2b+1) < a$ once $h < (a - b^2)/(2b+1)$, so $b + h \in S$, a contradiction. If $b^2 > a$, for $0 < h < b$, $(b - h)^2 > b^2 - 2bh > a$ once $h < (b^2 - a)/2b$; then $b - h$ is an upper bound of S (if $x \ge b - h$ then $x^2 \ge (b - h)^2 > a$), a contradiction. So $b^2 = a$. ■

In particular $\mathbb{Q}$ does not satisfy (C): $\{x \in \mathbb{Q} : x > 0, x^2 < 2\}$ has no rational supremum, as $\sqrt{2}$ is irrational.

Theorem 1.10 (1.2.7, uniqueness of the real field)

Let F and G be ordered fields satisfying the completeness axiom. There is a bijection $f : F \to G$ with $f(a + b) = f(a) + f(b)$, $f(ab) = f(a)f(b)$, and $a < b \iff f(a) < f(b)$.

Proof. Each ordered field contains a copy of $\mathbb{Q}$: $n \mapsto 1 + \cdots + 1$, extended to $\mathbb{Z}$ and to quotients; this map $\mathbb{Q} \to F$ preserves operations and order (Proposition 1.2). Let $\phi : \mathbb{Q}_F \to \mathbb{Q}_G$ be the resulting isomorphism of the rational subfields. Both F and G are Archimedean (Corollary 1.7), so the rationals are dense in each (Theorem 1.8). Define

$$f(x) = \sup\{\phi(r) : r \in \mathbb{Q}_F,\ r < x\}.$$

The set is nonempty and bounded above (by $\phi(s)$ for any rational $s > x$), so $f(x)$ exists. Order: if $x < y$, pick rationals $x < r < s < y$; then $f(x) \le \phi(r) < \phi(s) \le f(y)$. Also $f(r) = \phi(r)$ for rational r (density). So f is strictly increasing, hence injective. Surjective: for $z \in G$ let $x = \sup\{r \in \mathbb{Q}_F : \phi(r) < z\}$; density in G and strict monotonicity show $f(x) = z$. Additivity: for rationals $r < x$, $s < y$ we have $r + s < x + y$, so $\phi(r) + \phi(s) \le f(x + y)$, and taking suprema $f(x) + f(y) \le f(x + y)$; applying this to $-x, -y$ (note $f(-x) = -f(x)$, from the definition and density) gives equality. Multiplicativity for $x, y > 0$: products rs of positive rationals $r < x$, $s < y$ are exactly cofinal among rationals below xy, giving $f(xy) = f(x)f(y)$; extend to all signs using $f(-x) = -f(x)$. ∎

So the axioms determine $\mathbb{R}$ up to isomorphism.

Proposition 1.11 (1.2.8)

For nonempty bounded $S, T \subset \mathbb{R}$, $\sup(S+T) = \sup S + \sup T$ and $\inf(S+T) = \inf S + \inf T$, where $S + T = \{s + t\}$. If S, T consist of positive numbers, $\sup(ST) = \sup S \cdot \sup T$ and $\inf(ST) = \inf S \cdot \inf T$.

Proof. $\sup S + \sup T$ is an upper bound of $S+T$. Given $\epsilon > 0$ choose $s > \sup S - \epsilon/2$, $t > \sup T - \epsilon/2$; then $s + t > \sup S + \sup T - \epsilon$. For products, $\sup S \sup T$ is an upper bound of ST; with $a = \sup S$, $b = \sup T$ and $0 < \delta < \min(a, b)$, choose $s > a - \delta$, $t > b - \delta$, so $st > (a-\delta)(b-\delta) = ab - \delta(a+b-\delta)$, which is arbitrarily close to ab. For infima, $\inf S \inf T$ is a lower bound of ST, and with $s < \inf S + \delta$, $t < \inf T + \delta$ we get $st < \inf S \inf T + \delta(\inf S + \inf T + \delta)$. ∎

1.3 Limits of sequences

Motivation. Limits are the basic operation of analysis, and the ε–N definition turns "gets arbitrarily close" into something we can prove things about. The limit laws below let us compute limits without returning to ε each time.

(x_n) **converges** to x, $\lim x_n = x$, if for every $\epsilon > 0$ there is N with $|x_n - x| < \epsilon$ for all $n \ge N$.

Proposition 1.12 (1.3.1)

Limits are unique.

Proof. If $x_n \to x$, $x_n \to y$ and $x \ne y$, take $\epsilon = |x - y|/2$ and $n \ge \max(N_1, N_2)$: $|x - y| \le |x - x_n| + |x_n - y| < 2\epsilon = |x - y|$. ∎

Proposition 1.13 (1.3.2)

Every convergent sequence is bounded.

Proof. If $x_n \to x$, choose N with $|x_n - x| < 1$ for $n \ge N$; then $|x_n| \le \max\{|x_1|, \ldots, |x_{N-1}|, |x| + 1\}$. ∎

Theorem 1.14 (1.3.3, limit laws)

If $x_n \to x$ and $y_n \to y$, then (a) $x_n + y_n \to x + y$;
(b) $ax_n \to ax$;
(c) $x_n y_n \to xy$;
(d) if $y \neq 0$, then $y_n \neq 0$ for large n and $x_n/y_n \to x/y$;
(e) if $x_n \leq y_n$ for all large n, then $x \leq y$;
(f) (squeeze) if $x_n \leq z_n \leq y_n$ and $x = y$, then $z_n \to x$.

Proof. (a) $|(x_n + y_n) - (x + y)| \leq |x_n - x| + |y_n - y|$; use $\epsilon/2$ for each.

(b) is a special case of (c).

(c) With M bounding $|x_n|$ (Proposition 1.13), $|x_n y_n - xy| \leq |x_n||y_n - y| + |y||x_n - x| \leq M|y_n - y| + |y||x_n - x|$.

(d) For large n, $|y_n| > |y|/2$, and $|1/y_n - 1/y| = |y - y_n|/(|y_n||y|) \leq 2|y_n - y|/|y|^2$; combine with (c).

(e) If $x > y$, take $\epsilon = (x - y)/2$; for large n, $x_n > x - \epsilon = y + \epsilon > y_n$.

(f) For large n, $x - \epsilon < x_n \leq z_n \leq y_n < x + \epsilon$. ■

Example 1.1

(a) $1/n \to 0$ by Corollary 1.6.
(b) For $|r| < 1$, $r^n \to 0$: write $1/|r| = 1 + h$, $h > 0$; then $(1 + h)^n \geq 1 + nh$ (Bernoulli), so $|r|^n \leq 1/(1 + nh)$.
(c) $n^{1/n} \to 1$: with $n^{1/n} = 1 + h_n$, $n \geq \binom{n}{2} h_n^2$, so $0 \leq h_n \leq \sqrt{2/(n-1)}$.

1.4 Monotone sequences, limsup and liminf

Motivation. To prove a sequence converges without knowing its limit, we need intrinsic criteria. Monotone bounded sequences converge (this is completeness), and lim sup and lim inf always exist, measuring how far a sequence is from converging.

Theorem 1.15 (1.4.1, monotone convergence)

A bounded monotone sequence of real numbers converges.

Proof. Let (x_n) be increasing and bounded, and $x = \sup\{x_n\}$ (axiom (C)). Given ϵ, some $x_N > x - \epsilon$, and then $x - \epsilon < x_N \leq x_n \leq x$ for $n \geq N$. For decreasing sequences use inf. ■

Corollary 1.16 (1.4.2)

A series of nonnegative terms whose partial sums are bounded converges.

Proof. The partial sums form a bounded increasing sequence. ■

Example 1.2

$(1 + 1/n)^n$ is increasing and bounded by 3 (binomial expansion), so it converges to a number e. $\sum 1/n^p$ converges for $p > 1$: grouping terms in blocks $2^k \leq n < 2^{k+1}$, the partial sums are at most $\sum_k 2^{k(1-p)} < \infty$. $\sum 1/n$ diverges: the block sums are at least $\frac{1}{2}$.

For a bounded sequence (x_n), the sequences $\sup_{k\geq n} x_k$ and $\inf_{k\geq n} x_k$ are decreasing and increasing, so they converge (Theorem 1.15); define

$$\limsup_n x_n = \lim_n \sup_{k\geq n} x_k = \inf_n \sup_{k\geq n} x_k, \qquad \liminf_n x_n = \lim_n \inf_{k\geq n} x_k = \sup_n \inf_{k\geq n} x_k.$$

Clearly $\liminf x_n \leq \limsup x_n$, and $\liminf x_n = -\limsup(-x_n)$.

Proposition 1.17 (1.4.3)

For bounded (x_n), $a = \limsup x_n$ iff (a) for every $\epsilon > 0$ there is N with $x_n < a + \epsilon$ for all $n \geq N$, and (b) for every $\epsilon > 0$, $x_n > a - \epsilon$ for infinitely many n.

Proof. Let $s_n = \sup_{k\geq n} x_k \downarrow a$.

(a) Choose N with $s_N < a + \epsilon$; then $x_n \leq s_N$ for $n \geq N$.

(b) For each N, $s_N \geq a > a - \epsilon$, so some $k \geq N$ has $x_k > a - \epsilon$. Conversely, if a satisfies (a) and (b), then by (a) $s_n \leq a + \epsilon$ for large n, so $\limsup \leq a + \epsilon$; by (b) $s_n > a - \epsilon$ for all n, so $\limsup \geq a - \epsilon$. As ϵ is arbitrary, $\limsup x_n = a$. ■

Proposition 1.18 (1.4.4)

For bounded (x_n), $a = \liminf x_n$ iff (a) for every $\epsilon > 0$ there is N with $x_n > a - \epsilon$ for all $n \geq N$, and (b) for every $\epsilon > 0$, $x_n < a + \epsilon$ for infinitely many n.

Proof. Apply Proposition 1.17 to $(-x_n)$. ■

Theorem 1.19 (1.4.5)

If $x_n \to a$, then $\limsup x_n = \liminf x_n = a$. Conversely, if (x_n) is bounded and $\limsup x_n = \liminf x_n = a$, then $x_n \to a$.

Proof. If $x_n \to a$, then a satisfies conditions (a) and (b) of both propositions. Conversely, part (a) of Propositions 1.17 and 1.18 give $a - \epsilon < x_n < a + \epsilon$ for all large n. ■

Theorem 1.20 (1.4.6, nested interval theorem)

If $I_n = [a_n, b_n]$ are bounded closed intervals with $I_{n+1} \subset I_n$ for all n, then $\bigcap_n I_n \neq \emptyset$. If moreover $b_n - a_n \to 0$, the intersection is a single point.

Proof. (a_n) is increasing and bounded by b_1, so $a_n \to a = \sup a_n$; for all m, n, $a_n \leq a_{\max(m,n)} \leq b_{\max(m,n)} \leq b_m$, so $a \leq b_m$. Thus $a_m \leq a \leq b_m$ for all m. If x, y both lie in every I_n, $|x - y| \leq b_n - a_n \to 0$. ■

Conversely, the nested interval theorem together with the Archimedean property implies the completeness axiom: for S bounded above, bisect an interval $[s_0, u_0]$ ($s_0 \in S$, u_0 an upper bound) repeatedly, keeping the half whose right end point is an upper bound and whose interval meets S; the common point of the intervals is $\sup S$. Similarly, Theorem 1.15 is equivalent to (C).

1.5 Countable and uncountable sets

Motivation. $\mathbb{Q}$ and $\mathbb{R}$ are both dense, but there are "more" reals than rationals. Countability gives a precise way to say this and is used, for example, to show that discontinuities of monotone functions are rare.

A set is **countable** if it is finite or in bijection with $\mathbb{N}$.

Proposition 1.21 (countable unions)

Subsets of countable sets are countable; $\mathbb{N} \times \mathbb{N}$ is countable; a countable union of countable sets is countable. Hence $\mathbb{Z}$ and $\mathbb{Q}$ are countable.

Proof. An infinite $A \subset \mathbb{N}$ is enumerated increasingly by recursion ($a_1 = \min A$, $a_{k+1} = \min(A - \{a_1, \ldots, a_k\})$). $(m, n) \mapsto 2^m 3^n$ injects $\mathbb{N}\times\mathbb{N}$ into $\mathbb{N}$. If A_k are countable with surjections $f_k : \mathbb{N} \to A_k$ (for nonempty A_k), then $(k, n) \mapsto f_k(n)$ maps $\mathbb{N} \times \mathbb{N}$ onto $\bigcup A_k$, and a set that is the image of a countable set is countable (choose least preimages). $\mathbb{Q}$ is the image of $\mathbb{Z} \times \mathbb{N}$ under $(p, q) \mapsto p/q$. ■

Theorem 1.22 (Cantor)

$\mathbb{R}$ is uncountable.

Proof. Let $x_1, x_2, \ldots$ be any sequence of reals. Choose a closed interval I_1 of positive length not containing x_1, then $I_2 \subset I_1$ not containing x_2, and so on. By Theorem 1.20 some $x \in \bigcap I_n$, and $x \neq x_n$ for every n. ■

Consequently the irrational numbers are uncountable, and every interval of positive length is uncountable.